Exercise Week 7

Advanced Applied Statistics

Fall 2025

Problem 1: Jackknife Estimation*

In theis problem we use the jackknife method to estimate the standard error of the sample median. Let the observed data be $\mathcal{X} = \{12,\ 15,\ 13,\ 14,\ 16\}$.

(a) Compute the sample median.

(b) Apply the leave-one-out jackknife procedure to estimate the standard error of the sample median.

(c) Write the jackknife estimate of the bias.

(d) Comment on the stability of the median based on the jackknife samples.

Problem 2: Nonparametric Bootstrap*

Let us then use the bootstrap method to estimate the standard error and confidence interval of the sample mean. Use the same data: $\mathcal{X} = \{12,\ 15,\ 13,\ 14,\ 16\}$.

(a) Generate 1000 bootstrap samples from $\mathcal{X}$.

(b) Compute the sample mean for each bootstrap sample.

(c) Estimate the bootstrap standard error of the mean and the 95% bootstrap percentile confidence interval

(d) Plot a histogram of the bootstrap means.

Problem 3: Parametric Bootstrap*

Now, use the parametric bootstrap to estimate the standard error of the sample variance, assuming the data comes from a normal distribution. Again, use the data $\mathcal{X} = \{12,\ 15,\ 13,\ 14,\ 16\}$

(a) Assume the data comes from a normal distribution $N(\mu, \sigma^2)$. Estimate μ and σ from the sample.

(b) Generate 1000 parametric bootstrap samples from the fitted normal model.

(c) For each bootstrap sample, compute the sample variance.

(d) Estimate the standard error of the sample variance and a 95% confidence interval for the variance

(e) Compare the results to those from a nonparametric bootstrap (optional).

Problem 4: Influence Functions*

Let us investigate the influence function of the sample mean and sample median.

(a) Consider the sample $\mathcal{X} = \{1,\ 2,\ 3,\ 4,\ 100\}$

(b) Compute the sample mean and the sample median

(c) Add a new outlier value $x = 1000$ to the dataset, creating X'. Recompute the new sample mean and median. Which statistic changed more?

(d) What does this say about sensitivity to outliers?

(e) Define the influence function in words. Which estimator (mean or median) has a bounded influence function?

Problem 5: Additional Questions*

(a) What assumptions do each resampling method rely on?

(b) How does sample size affect jackknife and bootstrap accuracy?

(c) How can the influence function be used to detect outliers?